Part 1

Water Bottle - Version 1
Input: Write a product description for a water bottle
Output: # Stay Hydrated, Stay Refreshed

Meet your new everyday hydration companion! Designed for convenience, durability, and style, this water bottle makes staying hydrated easier wherever life takes you.

Whether you're heading to the gym, commuting to work, exploring the outdoors, or simply relaxing at home, its practical design makes it easy to enjoy your favorite beverages throughout the day.

Key Features:
- **Durable Construction:** Built for everyday use and long-lasting performance.
- **Portable Design:** Lightweight and easy to carry wherever you go.
- **Convenient to Use:** Easy to fill, drink from, and clean.
- **Versatile:** Perfect for water and your favorite cold beverages.
- **Eco-Friendly Choice:** A reusable alternative to disposable plastic bottles.

Make hydration part of your daily routine. Simple, practical, and stylish—this water bottle is ready to go wherever you do!

Read the output critically. What feels generic? What information is missing? Who does it seem to be written for?
The product description doesn't really get into anything specific about the product. For example, it doesn't talk about the material the product is made of, how much water the water bottle can hold, etc. It is written in an advertisement style where you might see in an amazon product description.

Part 2

Water Bottle - Version 2(Role Prompting)
Input: You are a product description expert. Write a product description for a water bottle.
Output: # Premium Stainless Steel Water Bottle

Stay refreshed wherever life takes you with our **Premium Stainless Steel Water Bottle**. Designed for everyday convenience and built to last, this stylish and durable bottle keeps your favorite beverages at the perfect temperature while complementing your active lifestyle.

Featuring double-wall vacuum insulation, it keeps drinks cold for up to 24 hours or hot for up to 12 hours. Its sleek, leak-proof design makes it easy to carry in your backpack, gym bag, or car, so you can enjoy refreshing hydration at home, at work, or on the go.

Made from high-quality, BPA-free materials, this reusable water bottle offers a practical and eco-friendly alternative to disposable plastic bottles.

Key Features
- **Temperature Control:** Double-wall insulation keeps drinks cold for up to 24 hours and hot for up to 12 hours.
- **Leak-Proof Design:** Secure lid helps prevent spills and leaks.
- **Premium Construction:** Durable stainless steel exterior designed for everyday use.
- **Safe & Reusable:** BPA-free materials for worry-free hydration.
- **Portable & Stylish:** Lightweight, sleek design fits most active lifestyles.
- **Eco-Friendly Choice:** Reduce single-use plastic waste with every refill.

Hydration made simple. Style made effortless. Take your favorite drink wherever the day leads.

Water Bottle - Version 3(Context)

Input: You are writing the product description of a brand new water bottle made by Apple. The water bottle comes in 3 variations, the iBottle, iBottle Pro, and iBottle Pro Max. All iBottle comes with Apple's temperature retention technology with iBottle holding up to 48 hours, iBottle Pro up to 72 hours, and iBottle Pro Max up to 90 hours.

Output:

Introducing iBottle. A new way to experience hydration.



Meet iBottle — hydration, intelligently reimagined.

Introducing Apple's first-ever smartly designed water bottle, engineered to keep your beverages at the perfect temperature for longer. Combining Apple's signature minimalist design with innovative temperature retention technology, iBottle makes every sip feel just right.

Whether you're heading to the office, hitting the gym, or exploring the outdoors, there's an iBottle designed for your lifestyle.

Three models. One extraordinary experience.



iBottle

48 Hours

The Essential

Everyday hydration, elevated. Enjoy Apple's temperature retention technology, keeping your beverages at your desired temperature for up to 48 hours.



iBottle Pro

72 Hours

The Professional

Go further with enhanced temperature retention. With up to 72 hours of temperature performance, iBottle Pro is built for those who expect more.



iBottle Pro Max

90 Hours

The Ultimate

The ultimate hydration experience. Featuring Apple's most advanced temperature retention technology, iBottle Pro Max keeps your beverages at the perfect temperature for up to an incredible 90 hours.

Designed to keep up with you.

- **Apple Temperature Retention Technology:** Advanced insulation designed to maintain your beverage's temperature for extended periods.
- **All-Day Hydration:** From your morning coffee to your evening workout, enjoy your favorite beverages just the way you like them.
- **Three Levels of Performance:** Choose between iBottle, iBottle Pro, and iBottle Pro Max.
- **Apple-Inspired Design:** A clean, sophisticated aesthetic that fits seamlessly into your everyday life.

The iBottle lineup at a glance

Feature	iBottle	Pro	Pro Max
Temperature retention	48 hrs	72 hrs	90 hrs
Technology	Apple	Apple	Apple
Experience	Essential	Advanced	Ultimate



Product concept copy: Apple branding and the described iBottle products and specifications are hypothetical, not verified Apple products.

Water Bottle - Version 4(Constraint)

Input: Write a product description for a water bottle. Please make sure that keep it concise and avoid using complicated jargons. Try to keep it under 300 characters and easy to read so that even a 5th grader will understand what they are reading.

Output: Stay refreshed wherever you go! This durable water bottle keeps your drinks cold or warm for hours. Its easy-to-carry design makes it perfect for school, sports, travel, and everyday use. Just fill it up and enjoy!

Water Bottle - Version 5(Few-shot)
Input: You are writing the product description of a water bottle. Here is an example of how I want the product description to look like for a portable induction cooktop:

About this item

- HIGH EFFICIENTY AND NO HOT KITCHEN - ChangBERT induction cooktop offers 9 preset power levels (200-1800W) and 18 preset temperature settings (120-460℉). transfers energy directly to magnetic cookware, heating the cookware itself rather than the surface for up to 83% energy efficiency. This direct heating method is more efficient than traditional gas or electric stoves, delivering instant response and stable performance, providing quick heat-up under high-demand kitchen environments.
- CONSISTENTLY STABLE HIGH-POWER WITH ONE-CLICK MAX 1800W POWER BUTTON - This induction burner built for constant use, it features continuous-duty circuitry for non-stop cooking, including an upgraded control board with dual IGBT. The unique power control technology with one-click max power button ensures fast and stable high-power output, reduces the temperature of internal electronics, and facilitates low-noise heat dissipation. This dependable performance sets it apart from others.
- LARGE COOKING AREA WITH 8-INCH HEATING COIL ENSURES PRECISE TEMPERATURE CONTROL FOR EVEN COOKING - Features an expanded 8” magnetic coil design that optimizes the winding structure to enhance the transfer of electromagnetic energy directly to the cookware, improving heating efficiency and ensuring more even heat distribution across the bottom of the cookware. This precise temperature control maintains steady heat to handling large cookwares for both professional kitchens and daily cooking.
- NSF CERTIFIED & COMMERCIAL GRADE DURABILITY - UL commercial standard 197, completely ETL & NSF approved. It meets industry durability standards, passing a drop test with an impact of 1.5 ft lbs. This cooker designed for large-volume usage, it is ideal for commercial and professional settings such as restaurants, hotel buffets, food trucks, institutional kitchens and catering services, while also perfect for large-volume usage.
- ENHANCED VERSION FOR BETTER PROFESSIONAL & CONVENIENT COOKING EXPERIENCE - Enhanced version equipped dedicated one-click buttons allow instant access to key functions, reducing steps during busy cooking. The extended 12-hour timer supports long simmering and unattended cooking with confidence. A large dual digital display clearly shows power or temperature and timer at the same time, easy to read in all lighting conditions and angles. Every upgrade delivers smoother operation for daily cooking.
- EASE OF USE - The physical control buttons with solid tactile response make operation intuitive and precise, reducing accidental touches during cooking, even with gloves or wet hands. With simple and clear display, the Key-Press-Control Panel provides clear visibility and easy to learn with no training needed. BOIL and KEEP WARM buttons let you jump straight to max power for boiling water or switch to steady holding without adjustments, perfect for fast heating and keeping food ready to serve.

- EASY TO CLEAN - Perfect for frequent, everyday use, this hot plate features a stainless steel construction and a full-glass surface that wipes clean in seconds and stays looking new over time. The fully sealed control panel keeps spills from getting inside, while the sleek, flush design means no hidden corners to scrub. Compared to traditional stoves, infrared burners, cleanup is quicker and easier, spend less time cleaning and more time enjoying your meals.
- STAINLESS STEEL STURDY CONSTRUCTION SUPPORTS UP TO 100 LBS - Made of premium stainless steel, this cooktop is built for long-lasting durability. The heavy-duty metal construction withstands high heat, humidity, and daily use without melting, cracking over time, delivering far greater stability than plastic housings when used with large or heavy cookware. Combined with a shatter-proof glass surface, both materials are easy to clean and looking new through frequent home and professional use.
- PORTABLE & COMPACT FOR EASY COOKING ANYWHERE - Engineered with high-quality materials and designed for effortless portability, this electric burner's compact design saves valuable counter space and is convenient for storage. For home use, it's perfect in home kitchens, dorms, apartments. For outdoor cooking, take it to RV trips, campsites, backyards, camping, cabins, boat, etc. and pair it with a portable power station delivering fast, even heating even with limited power.
- MULTIPLE SAFETY PROTECTIONS WITH 2 YEARS WARRANTY & SERVICE - Equipped with built-in auto-pan detection, over-heat protection, hot surface indication, small article detection, auto shutdown protection, and circuit failure protection. With no hot coils or open flames, it significantly reduces the risk of burns. 2-year limited warranty for home use and 1-year limited warranty for commercial use, with dedicated technical support. Contact us by email for more details.

Please use this example as reference for writing the product description of a water bottle.

Output: # About this item

- ADVANCED TEMPERATURE RETENTION TECHNOLOGY – KEEPS DRINKS COLD OR HOT FOR HOURS - Designed with advanced double-wall vacuum insulation, this water bottle helps maintain your drink's temperature for extended periods. Whether you prefer ice-cold water on a hot summer day or warm coffee during a chilly morning, enjoy your favorite beverages at the temperature you love. Perfect for everyday hydration, travel, workouts, and outdoor adventures.

- PREMIUM STAINLESS STEEL CONSTRUCTION – BUILT FOR LONG-LASTING DURABILITY - Made with high-quality stainless steel, this water bottle is designed to handle everyday use without easily rusting, cracking, or wearing down. Its sturdy construction provides reliable performance at home, work, school, or outdoors. The durable exterior also helps protect the bottle from everyday bumps and scratches.

- LEAKPROOF DESIGN – DRINK WITH CONFIDENCE ON THE GO - Features a secure, leak-resistant lid that helps prevent unwanted spills and leaks. Toss it into your backpack, gym

bag, or car without worrying about water getting onto your belongings. The easy-to-open lid makes drinking simple and convenient, whether you're commuting, exercising, or traveling.

- LARGE CAPACITY – STAY HYDRATED THROUGHOUT THE DAY - Designed to hold plenty of water to keep you hydrated throughout your busy day. Spend less time refilling and more time enjoying your activities. Whether you're working at your desk, hitting the gym, going to school, or exploring the outdoors, this bottle makes it easy to keep your favorite drink close at hand.

- WIDE MOUTH DESIGN – EASY TO FILL AND CLEAN - The wide opening makes filling your bottle with water, ice cubes, and your favorite beverages quick and easy. It also makes cleaning the inside simple, helping you keep your bottle fresh and ready for everyday use. No more struggling to fit ice cubes or reaching into narrow openings.

- COMFORTABLE AND PORTABLE – PERFECT FOR EVERYDAY USE - Designed with a lightweight, easy-to-carry body that fits comfortably in your hand and most standard cup holders. Its compact design makes it convenient to take anywhere, including the office, gym, school, camping trips, road trips, and outdoor adventures. Stay hydrated wherever life takes you.

- SWEAT-FREE EXTERIOR – KEEPS YOUR HANDS AND BAGS DRY - The double-wall insulation helps prevent condensation from forming on the outside of the bottle. No more wet hands, water rings on your desk, or moisture inside your backpack. Enjoy a comfortable grip while keeping your surroundings clean and dry.

- EASY TO USE AND MAINTAIN – MADE FOR DAILY HYDRATION - Simple to open, fill, drink from, and clean. The convenient lid design makes it easy to enjoy your beverages without unnecessary hassle. Built for everyday convenience, this water bottle is a reliable choice for busy mornings, long workdays, workouts, and weekend adventures.

- VERSATILE DESIGN – YOUR PERFECT HYDRATION COMPANION - Whether you're at home, in the office, at school, at the gym, or enjoying the outdoors, this water bottle is ready for the job. Its combination of temperature retention, durable construction, and convenient portability makes it a practical choice for anyone looking to stay hydrated throughout the day.

Note: Since you haven't provided specific bottle specifications yet, I've kept the description general and avoided making unsupported claims about capacity, insulation duration, or certifications.

I can also adapt this into an Amazon-style listing with stronger selling points and SEO-friendly keywords.

Water Bottle - Version 6(Output Format)
Input: Write a product description for a water bottle. Please include an overview of the product then provide bullet points for the features of the water bottle.
Output: # Water Bottle – Stay Hydrated Anywhere

Stay refreshed wherever life takes you with this durable and easy-to-use water bottle. Designed for everyday use, it keeps your drinks at the right temperature while making it easy to stay hydrated at school, work, the gym, or on the go.

Key Features

- **TEMPERATURE RETENTION** – Keeps drinks cold or hot for hours, so you can enjoy your favorite beverages at the perfect temperature.
- **DURABLE DESIGN** – Made with high-quality materials to handle everyday use.
- **LEAKPROOF LID** – Prevents spills and leaks, making it safe to carry in your backpack or bag.
- **EASY TO CARRY** – Lightweight and portable, perfect for travel, school, work, and outdoor activities.
- **EASY TO CLEAN** – Simple design makes cleaning quick and hassle-free.
- **REUSABLE AND ECO-FRIENDLY** – Helps reduce the need for single-use plastic bottles while keeping you hydrated.

Part 3

Noise-canceling headphone

Input: You are an audiophile who has years of experience reviewing headphones. You are writing the product description of a brand new Noise-canceling headphone from a tech company.

Here are the main features of the product:

- Brand new state of the art ANC
- Over 100 hours of battery with ANC
- Sleek ergonomic design
- Lossless audio
- Spatial audio

Audience: Everyday consumers but the product is also good enough to satisfy other audiophiles.

Here is an example of a product description of a gaming pc: Level up your setup with the iBUYPOWER EBI7N5704, the perfect gaming PC build for someone looking to upgrade their desktop computer system for video games, editing, and streaming. This prebuilt iBUYPOWER Gaming PC utilizes professionally vetted components to create a high-performing desktop machine. Take gaming and content creation to a new level on an Intel Core i7 14700F processor paired with an NVIDIA GeForce RTX 5070 12GB graphics card. Enjoy fast load times and ample storage space on a 1TB NVMe solid-state drive. Multitask like no other thanks to 32GB DDR5 RGB RAM. The PC is even Wi-Fi Ready and comes with Windows 11 Home installed so you can plug in and game right away. Discover the gamer experience and upgrade your RGB setup thanks to the included iBUYPOWER gaming keyboard and mouse. To keep your computer cool and quiet at full power, the EBI7N5704 boasts the new Tower Air Cooler, a

performance air fan CPU cooler, and RGB fans in a premium iBUYPOWER Element case.

You can use the product description of the gaming pc as a reference for writing the product description of the Noise-canceling headphone.
Output: Take your listening experience to the next level with our brand-new noise-canceling headphones, designed for music lovers, everyday listeners, and audiophiles alike. Featuring state-of-the-art Active Noise Cancellation (ANC), these headphones block out unwanted background noise so you can focus on what matters most: your music. Enjoy your favorite songs, podcasts, and movies without worrying about frequent charging, thanks to an incredible battery life of over 100 hours, even with ANC turned on. Experience every note and detail with lossless audio, delivering rich, clear, and high-quality sound that brings your music to life. Take your entertainment even further with immersive spatial audio that makes you feel like you're right in the middle of the action. All of this comes in a sleek, ergonomic design built for lasting comfort, making these headphones the perfect companion for long listening sessions, daily commutes, travel, and everything in between. Whether you're relaxing at home or on the move, enjoy exceptional sound, powerful noise cancellation, and comfort without compromise.

Reusable grocery bag
Input: You are a product advertiser who is in charge of writing the product description of a reusable grocery bag.

Here are the main features of the product:
- Foldable for easy storing
- Insulated for keeping frozen food frozen for up to 4 hours
- Comes in a variety of different designs and colors

Audience: Everyday families

Rules:
- Keep it under 100 words
- Use upbeat tone
- Place an emphasis on the insulation feature

Output: Make grocery shopping easier and more enjoyable with our versatile insulated reusable grocery bag! Designed for busy families, it folds flat for easy storage and keeps frozen foods cold for up to 4 hours, so your ice cream, frozen meals, and other groceries stay fresh on the way home. Available in a variety of fun colors and stylish designs, it's the perfect everyday shopping companion!

Analysis:
I tried using role prompting, context, constraints, few-shot prompting, and output format for the product description of a water bottle with each technique used in a separate chat log. For the product description of a Noise-canceling headphone, I incorporated role prompting, context, and

few-shot prompting together. For reusable grocery bags, I incorporated role prompting, context, and constraints together.

For role prompting, I noticed that depending on the role you define, the output will incorporate new language and details that may not appear if you didn't define a role. This can be seen in Water bottle - Version 2 where the output included details such as the material of the bottle and how long it can keep drinks cold/hot for. These details did not appear in Water bottle - Version 1. Adding context to the prompt provided more specific details that the LLM can use in its output when describing the product as seen in Water bottle - version 3 instead of the vague language of the water bottle's feature in version 1. Constraint I personally didn't find it too helpful as I feel like it made the output worse possibly due to the rules I set not being well defined enough(Version 4). Few-show prompting provided the LLM with an exact template to follow and the output was able to follow that template closely as seen in version 5. Finally, the output format I find to be similar to few-show prompting except that instead of showing the LLM examples of the output, we are describing to it what the output should look like which it was able to do in version 6.

One occurrence of a detail being made up was when I applied for a role prompting technical for the product description of a water bottle. Despite not providing any details about the features of the water bottle, the output made up its own features describing the water bottle. To tackle misinformed features, in the next water bottle version, I provided context of the exact features of the water bottle.

The technique that made the biggest difference is few-shot prompting due to the fact that it provides the LLM an concrete template to follow versus the other techniques that left some interpretation up to the LLM.